

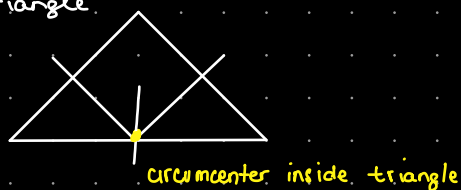
Theory 3

3.1.1 a)

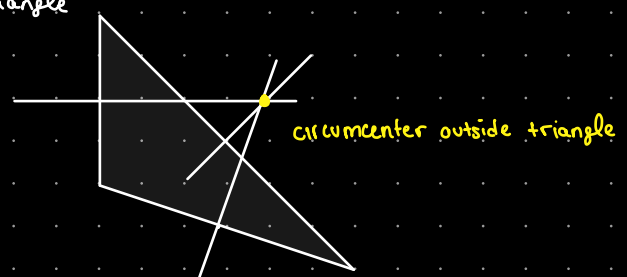
For obtuse angles the circumcenter of the triangle would be located outside the triangle itself, which would pull the Voronoi boundary outward (which creates spikes).

- to calculate Voronoi area, we use: $\cot()$
- $\cot(\theta) = \frac{1}{\tan(\theta)}$ and for $90^\circ < \theta < 180^\circ$ ist $\tan(\theta) < 0$
- this means: for obtuse angles the area could be negative

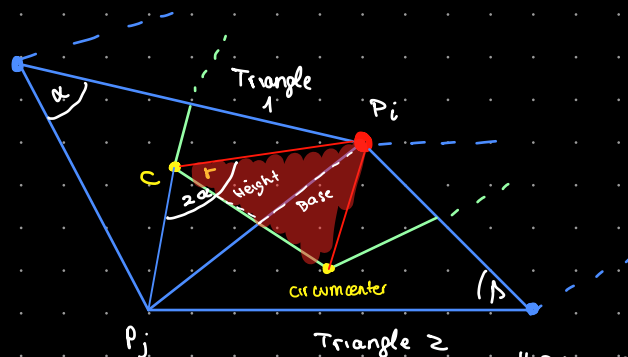
acute triangle



obtuse triangle



b) Area of the red triangle



1. The Base b is the edge between p_i and p_j : $\frac{\|p_j - p_i\|}{2}$

2. The Height of the red triangle is from the middle from the base to the circumcenter.
- this can be calculated using $\tan()$
 - the angle of the triangle formed by p_i, p_j, c is $2 \cdot \alpha$
 - so the angle of the red triangle: $\tau = \alpha$

Height h : $\tan(\alpha) = \frac{b}{h} \Leftrightarrow h = \frac{b}{\tan(\alpha)}$

• with : $\frac{1}{\tan(\alpha)} = \cot(\alpha)$

$\Rightarrow h = b \cdot \cot(\alpha)$

$\Leftrightarrow h = \frac{\|p_j - p_i\|}{2} \cdot \cot(\alpha)$

Area for red triangle of Triangle 1

$$A_1 = \frac{h \cdot b}{2}$$

$$\Leftrightarrow \frac{\|p_j - p_i\|}{2} \cdot \cot(\alpha) \cdot \frac{\|p_j - p_i\|}{2} \cdot \frac{1}{2}$$

$$\Leftrightarrow \frac{1}{8} \cdot \cot(\alpha) \cdot \|p_j - p_i\|^2$$

• The same follows for the red area of Triangle 2

$$A_2 = \frac{1}{8} \cdot \cot(\beta) \cdot \|p_j - p_i\|^2$$

• Then sum both areas:

$$A = A_1 + A_2$$

$$\Leftrightarrow \frac{1}{8} \|p_j - p_i\|^2 (\cot(\alpha) + \cot(\beta))$$

• This is calculated for each edge with p_i ($\sum$ comes out of this)

$$\Rightarrow A(i) = \frac{1}{8} \sum_{j \in N_1(i)} (\cot(\alpha) + \cot(\beta)) \cdot \|p_j - p_i\|^2$$

3.1.2 a)

Memory for a triangle Mesh without boundaries:

$$\text{Mem} = v \cdot 10 \text{ Bytes} + \overset{e \approx 3v}{\uparrow} 3v \cdot 8 \text{ Bytes} + \overset{f \approx 2v}{\uparrow} 2v \cdot 8 \text{ Bytes} + \overset{\text{pro edge 2 Halfedges}}{3v \cdot 2 \cdot 32 \text{ Bytes}}$$

$$\Leftrightarrow v (10 \text{ Bytes} + 3 \cdot 8 \text{ Bytes} + 2 \cdot 8 \text{ Bytes} + 6 \cdot 32 \text{ Bytes})$$

$$\Leftrightarrow v \cdot 252 \text{ Bytes}$$

b)

Ratio for quad Meshes:

→ Two triangles create one quad and the diagonal edge disappears:

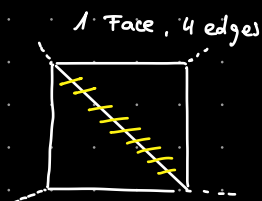
→ Two faces always share one edge, so: $e = 2f$

$$\Rightarrow v - e + f = \chi$$

$$\Leftrightarrow v - e + f \approx 0$$

$$\Leftrightarrow v - 2f + f \approx 0$$

$$\Leftrightarrow v \approx f$$



$$\text{Ratio } (v : e : f) = 1 : 2 : 1$$

c) Memory of quad Meshes:

$$e \approx 2v \quad f \approx v \quad \text{Half edges}$$

$$\text{Mem} = v \cdot 20 \text{ Bytes} + 2v \cdot 8 \text{ Bytes} + v \cdot 8 \text{ Bytes} + 2v \cdot 2 \cdot 32 \text{ Bytes}$$

$$\Rightarrow v \cdot 172 \text{ Bytes}$$